

Classifying Convex and Non-Convex Polygons

Jonathan R. Bacolod

A **polygon** is a closed plane figure formed by three or more **line segments** called **sides** that meet only at their **endpoints**, called **vertices**.

Convex Polygon

A **convex polygon** is a polygon in which every **interior angle** measures less than 180° .

A line segment connecting any two interior points lies entirely **inside** the polygon.

Non-Convex Polygon

A **non-convex polygon**, also called a **concave polygon**, has at least one **interior angle** greater than 180° .

This creates an inward **dent** or **indentation**.

Regular Polygons

All regular polygons are convex.

A regular polygon is both equilateral and equiangular.

Examples

Example 1

A regular triangle has all interior angles equal to 60° .

Since every angle is less than 180° , the triangle is **convex**.

Example 2

A hexagon has one interior angle measuring 210° .

Since one angle is greater than 180° , the hexagon is **non-convex**.

Example 3

An L-shaped hexagon has an inward indentation.

The indentation creates a reflex interior angle greater than 180° .

Therefore, the polygon is **non-convex**.

Practice Exercises

Practice 1

A quadrilateral has interior angles measuring 85° , 95° , 110° , and 70° .

Classify the polygon as **convex** or **non-convex**.

Practice 2

A pentagon has interior angles measuring 100° , 110° , 95° , 105° , and 130° .

Classify the polygon as **convex** or **non-convex**.

Practice 3

A hexagon has one interior angle measuring 210° and the rest less than 180° .

Classify the polygon as **convex** or **non-convex**.

Practice 4

A square has all sides equal and all angles equal to 90° .

Classify the polygon as **convex** or **non-convex**.

Practice 5

An L-shaped hexagon is formed by cutting a small rectangle from the corner of a larger rectangle.

Classify the polygon as **convex** or **non-convex**.

Practice 6

A regular octagon has all interior angles equal to 135° .

Classify the polygon as **convex** or **non-convex**.

Practice 7

A five-pointed star (pentagram) has angles alternating between acute and reflex measures. Classify the polygon as **convex** or **non-convex**.

Activities

Activity 1

A regular hexagon has all interior angles equal to 120° .

Classify the polygon as **convex** or **non-convex**.

Activity 2

A quadrilateral has interior angles measuring 72° , 108° , 93° , and 87° .

Classify the polygon as **convex** or **non-convex**.

Activity 3

A pentagon has interior angles measuring 88° , 120° , 102° , 115° , and 115° .

Classify the polygon as **convex** or **non-convex**.

Activity 4

A pentagon has one interior angle measuring 195° and the remaining angles less than 180° . Classify the polygon as **convex** or **non-convex**.

Activity 5

A rhombus has all sides equal and all angles equal to either 60° or 120° .

Classify the polygon as **convex** or **non-convex**.

Activity 6

A T-shaped octagon is formed by joining two rectangles such that one sits on top of the other at its center.

Classify the polygon as **convex** or **non-convex**.

Activity 7

A regular decagon has all interior angles equal to 144° .

Classify the polygon as **convex** or **non-convex**.

Activity 8

A six-pointed star has boundary angles alternating between acute and reflex measures. Classify the polygon as **convex** or **non-convex**.

Activity 9

A pentagon is shaped like a lightning bolt, with one interior angle measuring 230° .

Classify the polygon as **convex** or **non-convex**.

Activity 10

An octagon has all interior angles less than 180° and every diagonal contained entirely within the figure.

Classify the polygon as **convex** or **non-convex**.

Thank you!